

VIVA PRESENTATION & DEFENSE SCRIPT

Device Risk Analysis | Case Study No. 86

Candidate: Atharva Gahine | B.Tech CSE Semester V (2024–2028)

1. Oral Presentation Structure (10-Minute Guide)

Opening: Greet examiners, state project title, Case Study No. 86, and academic context.

Problem: Explain enterprise endpoint risk and manual SOC alert fatigue.

Data & Preprocessing: Detail 3,012 assets, deduplication, median/mode imputation, and feature engineering.

Model Benchmarks: Highlight 6 algorithms, 5-fold CV, and champion test accuracy of 89.00% (F1: 0.8898).

Demonstration: Walk examiners through Streamlit home KPIs, live prediction, and feature insights.

2. Top Technical Viva Questions & Crisp Answers

No.	Viva Question	Crisp Technical Answer
Q1	Why did Logistic Regression win?	Risk is an additive latent score; regularized linear models capture monotonic accumulation best.
Q2	How was data leakage avoided?	ColumnTransformer inside scikit-learn Pipeline objects fit strictly on training splits.
Q3	Why Weighted F1 over Accuracy?	Accounts for precision-recall balance and mitigates penalties for false negatives.
Q4	What does Attack Surface Index represent?	Consolidates perimeter open ports (0.4) and failed login brute-force attempts (0.6).
Q5	How do you handle unseen inputs in app?	OneHotEncoder(handle_unknown='ignore') and defensive try-catch loading blocks.